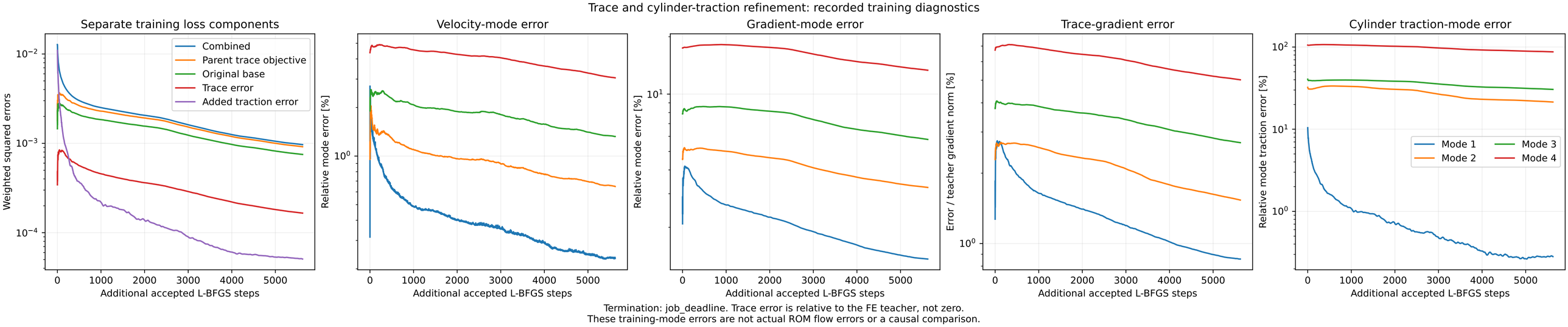

Trace and cylinder-traction refinement: recorded training diagnostics
Separate training loss components
Weighted squared errors
Combined
Parent trace objective
Original base
Trace error
Added traction error
Additional accepted L-BFGS steps
Velocity-mode error
Relative mode error [%]
Additional accepted L-BFGS steps
Gradient-mode error
Relative mode error [%]
Additional accepted L-BFGS steps
Trace-gradient error
Error / teacher gradient norm [%]
Additional accepted L-BFGS steps
Cylinder traction-mode error
Relative mode traction error [%]
Mode 1
Mode 2
Mode 3
Mode 4
Additional accepted L-BFGS steps
Termination: job_deadline. Trace error is relative to the FE teacher, not zero.
These training-mode errors are not actual ROM flow errors or a causal comparison.